DB4 Review Meeting 4 - 5 Minute Presentation Script

Final Product: automated mussel bioreactor with web control, pump calibration, PID cooling, and biological feeding/waste loop.

Section	Main message
Final product	Controlled system for supporting mussels with sensors, pumps, cooling, and monitoring.
Web server	Real-time dashboard for monitoring and manual control.
Pump calibration	Flow rate measurements convert required dose volume into pump run time.
PID temperature control	Cooling from 26 C to a stable 17-18 C mussel-safe range.
Biological model	Algae feeding and waste transfer create a practical closed-loop model.

Slide 1 - Final Product

Hello, today we are presenting our final DB4 product. Our project is an automated mussel bioreactor system. The goal was to create a small controlled environment where mussels can stay alive and healthy while we monitor and control important parameters such as temperature, algae feeding, waste transfer, and system operation.

The main idea is that the mussel tank should not just be a static container. It should behave like a controlled biological system, where the electronics, sensors, pumps, and web interface work together to support the living organism. The system includes a cooling loop, algae and waste pumps, temperature sensing, pump calibration, PID temperature control, and a biological model connecting algae feeding and mussel waste.

Slide 2 - Web Server

This slide shows our web server interface. The web server allows us to monitor the system in real time and manually control the main components. From the interface, we can see the current temperature, system status, and control outputs. We can also switch pumps and cooling functions on or off.

The reason this is important is that the system needs to be both autonomous and controllable. During testing, manual control helped us debug hardware problems, check pump behavior, and safely stop components if something was not working correctly. In the final setup, the web server acts as a simple monitoring and control dashboard for the bioreactor.

Slide 3 - Pump Calibration

This slide explains the pump calibration. We tested the pumps by running them for a fixed amount of time and measuring the collected volume. From this, we calculated the flow rate.

For the cooling pump, we tested different PWM values. At PWM 500, the flow rate was about 240.7 mL/min, and at PWM 700, it increased to about 422.7 mL/min. This shows that the cooling pump flow can be controlled by changing the PWM signal.

For the algae and waste pumps, we use the same pump type, so we use the same calibrated flow rate for both: Q = 0.5 mL/s = 30 mL/min. This is useful because the software can convert a required dose volume into a pump running time.

For example, if we want to pump 10 mL, and the pump gives 0.5 mL/s, then the pump should run for 10 / 0.5 = 20 seconds. So the pump calibration makes the biological dosing reproducible instead of random.

Q = 0.5 mL/s = 30 mL/min | t_pump = V_dose / Q | 10 mL / 0.5 mL/s = 20 s

Slide 4 - PID Temperature Control

This slide shows the PID temperature control result. The purpose of the PID controller is to keep the mussel tank temperature inside the healthy range. Mussels like approximately 17-18 C, so our setpoint was chosen as 17.75 C.

The experiment starts at 26 C, which is too warm for the mussels. The cooling system first cools the water down from 26 C toward the target range. After the temperature reaches around 18 C, the PID controller reduces the cooling action and maintains the temperature close to the setpoint.

The important result is that the system does not simply cool continuously. Instead, it reacts to the temperature error. If the temperature is too high, it increases cooling. If the temperature is close to the target, it reduces the output to avoid overshooting below 17 C.

From the plot, we can see that after the cooling phase, the temperature stays inside the ideal range of 17-18 C with only small fluctuations. This proves that the cooling system can maintain stable thermal conditions for the mussel experiment. This is one of the most important parts of the project because temperature control is essential for keeping the biological system alive.

Slide 5 - Biological Model: Feeding and Waste Loop

This slide explains the biological loop. The idea is to connect the algae tank and the mussel tank into a simple closed biological model.

First, the algae tank contains the algae used as food. The OD sensor estimates the algae concentration. Then the algae pump transfers a controlled amount of algae water into the mussel tank. The mussel filters the algae from the water as food. After filtration and digestion, the mussel produces waste. Then the waste pump transfers waste-rich water back toward the algae tank, where the waste can act as a nutrient input for algae regrowth.

So the loop is: algae tank to algae pump to mussel tank to mussel filtration and waste to waste pump and back to the algae tank. This creates a connection between feeding the mussel and feeding the algae. The system is not fully biologically closed yet, but the model gives us a structure for controlling the timing and volume of the two pumps.

The control equations are based on our pump calibration: Qalgae = Qwaste = 0.5 mL/s, and tpump = Vdose / Q. So for a 10 mL dose, both pumps run for 20 seconds.

We also used measured biological data in the model. The algae growth rate was estimated as mu = 0.035 h^-1, which gives a doubling time of about 19.7 hours. The mussel clearance rate was estimated as approximately 1.80 L/h. These values help us choose pump timing and understand how fast algae is consumed and regenerated.

Q_algae = Q_waste = 0.5 mL/s | t_pump = V_dose / Q

Slide 6 - Waste Pump Timing and OD Data

This slide shows the timing logic for the biological loop. At the start, the algae pump runs for 20 seconds, which gives approximately 10 mL of algae water to the mussel tank.

Then we wait about 60 minutes. This delay represents the time where the mussel filters the algae and the system allows digestion and waste production to happen.

After this delay, the waste pump runs for 20 seconds, transferring about 10 mL of waste-rich water back toward the algae tank.

So the timing is: at 0 minutes, algae pump runs for 20 seconds; at 60 minutes, mussel filtration and digestion delay; after 60 minutes, waste pump runs for 20 seconds. This makes the biological model practical in code because the pump actions are based on measured flow rates, not guesses.

We also found that the red and green OD channels showed the strongest concentration response, meaning they were more useful for estimating algae concentration than the other channels.

Closing conclusion

To conclude, we conducted four main parts in this final prototype: we built a real-time web interface, calibrated the pumps, implemented PID temperature control, and created a biological feeding and waste loop.

The most important result is that the system can maintain the mussel tank near the desired 17-18 C range, while the pump calibration allows controlled algae feeding and waste transfer. The biological model connects the mussel and algae tanks using measured flow rates, algae growth, and mussel clearance data.

Overall, the prototype shows that our system can support the living mussel environment with controlled temperature, reproducible dosing, and real-time monitoring. The next improvements would be more long-term biological testing, better OD calibration, and validation of the waste loop over several days.